

OOP Programming Assignment (Date: 21-April 2026)

1. Student Record System

- Create a class Student with data fields name, roll and marks for 3 subjects. The student data can be entered using
 - Constructors
 - Method in the class
- Methods in the class should display the data of the student. Calculate average marks and provide a grade depending on the average marks.
- Extend this class to create another class Group which dynamically adds new students in the Group. Implement the following methods:
 - Find the topper
 - Find the students getting F grade
 - Delete a student with name or roll matching

2. 3D Geometry

- Create a class called Point to store the coordinates of a point in 3D. Implement the following methods:
 - Distance between two points
 - Distance from origin
 - Parameters of a line passing through two points
- Now create a shape class for defining the dimensions of a 3D shape. Data fields will be center of the shape and length, width and height to store the dimensions. Implement the following methods to calculate:
 - Volume – should do some calculation
 - Surface area – should do some calculation
 - Virtual function: point lying inside the shape
- Three classes cuboid, cube and sphere from shape. No separate data fields to be stored in these. The methods volume, surface area should use the calculations in the base class function. Overwrite the function point lying inside the shape in all three classes.

3. Time Class

- Data fields to contain are hours, mins, secs and meridian (am or pm).
 - Use constructors and user defined data entry functions
 - Function to display time in both formats AM/PM or 24-hour format
 - Overload + operator to add two time objects
 - Overload + operator to add an integer (user to define whether the integer to be added is hrs, mins or secs)

4. User Defined String Class

- Create a class String:
 - Dynamically allocate memory for string
 - Implement:
 - Constructor
 - Copy constructor
 - Destructor

5. Shopping Cart System

- Create a product class with different products: two data-fields type, price
- Create a class called cart which dynamically adds and deletes item in the cart. It should have a function to display the bill for items stored in the cart.

6. Template Class

- Create a template class MathOperations
- Define static member functions for add, subtract, multiply, power and division functions. Ensure that the operations can be performed on different datatypes. Each operation will take two operands
- Can you overload static member functions. If yes, show it and if no why not.

7. Modify the String class of problem-4 to include a static data field which holds the length of the longest string entered. Using this static data field, design a manipulator to print all the objects right-aligned.

8. Write recursive functions for:

- Sum of digits
- Reverse number
- Check palindrome

9. Predict and explain the output:

```
int arr[]={1,2,3,4,5};  
  
int *p = arr;  
  
cout << *p++ << endl;  
  
cout << (*p)++ << endl;  
  
cout << *++p << endl;
```

10. Flatten 2D Array

- Convert 2D array into 1D using pointer logic